

exactly 80 mm: _____

FIRST SOLVE

Now here? Do the walkthrough once at cubepath3.svevel.app/learn/1. This is the memory jog.

NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube
 ' = counter-clockwise, 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together
 y = spin the whole cube left, white staying down. A whole-cube turn never noting.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run at as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the YELLOW face first, each about its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'U'R



white sticker FRONT
L'U'L



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow, turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U R'UR' y L'U'L



goes LEFT
U' L'U'L y' R'UR'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ RUR' sexy move
■ RUR' Sune
■ R'FRP' sledgehammer
gap = change grip

FIRST SOLVE

New here? Do the walkthrough once at [cubepath52.vercel.app/learn](#). This card is the memory jog.

NOTATION

R U F L D B = turn that face clockwise, look at it from outside the cube

' = counter-clockwise, 2 = half turn. Lowercase $r f l$ = that face plus the layer behind it, turning together

y = spin the whole cube left. Stay case down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top. Turn U until it sits above the corner slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'ULU



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: Turn U until its front color matches the center below it. The arrow shows which slot it goes to.

goes RIGHT
U' R'UR' y L'ULU



goes LEFT
U' L'ULU y' RUR'U'



Edge stuck in the wrong slot? Run either one to kick it out, then place it.

R U F L D B = turn face clockwise
r u f l d b = turn face counter-clockwise
2R 2U 2F 2L 2D 2B = half turn
R' U' F' L' D' B' = turn face counter-clockwise
y = spin the whole cube left

TWO-LOOK

OLL CROSS - yellow edges



Line bar left to right
FRUR'U'F'



Hook hook pointing front-right
fFRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RU'R'U' RU2R'



Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'



PI $\frac{2}{3}$ yellow; headlights left
fFRUR'U'f' FRUR'U'F'

OLL CORNERS - continued



Headlights $\frac{3}{4}$ yellow; headlights at the back
R2UR'U2 RD'R'U2R'



2x Headlights $\frac{3}{4}$ yellow; headlights both sides
RU'R'U' RU'R' URU2R'



Chameleon $\frac{3}{4}$ yellow next to each other
fFRUR'U' FR'



Bowtie $\frac{3}{4}$ yellow diagonal
fFRUR'U' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn around, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at the end? Run the first one, then read again.

■ **RU'R'U'** sexy move ■ **RU'R'U'U'** ■ **R'FR'** slide change ■ **g** = change grip

2-LOOK

OLL CROSS - yellow edges



Line has left-to-right
fRUR'U'f'

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ 1 yellow, rest clockwise
RUR'U' RU2R'

Anti-Sune $\frac{1}{2}$ 1 yellow, rest counter-clockwise
RU2R' U'RUR'

Pi $\frac{3}{4}$ 0 yellow, headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS - continued



Headlights $\frac{3}{4}$ 2 yellow, headlights at the back
RD2RU2R' RD'R'U2R'

2x Headlights $\frac{3}{4}$ 0 yellow, headlights both sides
RUR'U' RU'R' URU2R'

Chameleon $\frac{3}{4}$ 2 yellow next to each other
rUR'U'r' FRF'

Bowtie $\frac{3}{4}$ 2 yellow diagonal
F'rUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn applied, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sune and read again.

■ RUR'U' sexy move
 ■ RUR'U'
 ■ R'FRF' sledgehammer
 gap = change grip

ONE-LOOK PLT

ADJACENT CORNER SWAP

exactly one face shows headlights



• T1: headlights: pairs B-left F-left
RUR'U' R'F RZUR'UR' RUF'F'

Ja L solid: pairs B-right F-left B-front
x RZFRF' RU2 rUr U2

Jb L solid: pairs B-left F-back
RUR'F' RUR'U' R'F RZUR'U'

Aa L solid: pairs F-right B-front
x LZD2 L'UL D2 L'UL'

Ab L solid: pairs B-right B-back
x' LZD2 LUL' D2 LU'L

F1 L solid
RUF'F' RUR'U' R'F RZUR'UR' RUF'UR'

One face shows headlights: two corners of that face match. Hold as shown, turn, then turn top to finish.

■ RUF'F' seq.mvcs. ■ RUF'U' seq.mvcs. ■ R'FR'F' slideschamber. seq. = change dir.

ADJACENT CORNER SWAP

continued



Ra L1: headlights: pairs F-right
RU'R' UR' RU'R UR'D'RU'D' R'U2R'



Rb L1: headlights: pairs B-left
R2F RURUR' FR U2RU2UR



Ga L1: headlights: pairs F-right
RZUR'UR'U' RU'R2 U'D R'UR'D'



Gb L1: headlights: pairs B-back
RU'R UD' RZUR'UR'U' RU'R2D



Gc L1: headlights: pairs B-right
RZUR'UR'U' RU'R2 UD' RU'R'D



Gd L1: headlights: pairs B-front
RUR' UD' RZUR'UR'U' UR'UR2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights

• T1 headlights: pairs R-left F-left
RUR'U' R'F R2UR'U' RUR'F'

• Ja1 solid: pairs B-right F-left R-front
x R2FRF' RU2 rUr U2

• Ib1 solid: pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2UR'

• Aa1 headlights: pairs F-right B-front
x L2D2 L'U'L D2 L'U'L'

• Ab1 headlights: pairs B-right R-back
x' L2D2 LUL' D2 LU'L

• F1 solid
RU'F' RUR'U' R'F R2UR'U'

One face shows two headlights; two corners of that face match. Hold as shown, then turn the top to finish.

■ **RUR'U'** = **u** move ■ **RUR'U'** = **u** move ■ **R'FRF'** = **u** slicehammer **can** = **change grip**

ADJACENT CORNER SWAP

continued

• Ra1 headlights: pairs F-left R-left
RUR'U' RUR DR'UR'D' RU'2R'

• Rb1 headlights: pairs B-left R-back
R2F RURU' FR U2RU'2R

• Ga1 headlights: pairs F-right R-front
R2UR'UR'U' RU'R2 U'D R'UR'D'

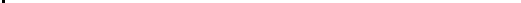
• Gb1 headlights: pairs B-back R-back
RUR' U'D' R2UR'URU' RU'R2D

• Gc1 headlights: pairs B-right R-front
R2UR'URU' RU'R2 U'D' RU'R'D

• Gd1 headlights: pairs R-front R-back
RUR' U'D' R2UR'URU' UR'UR'D

Learn in the printed order, three a week, about six weeks. Ja and Ib on the same day — one is the mirror of the other.

[illegible][illegible]

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm: 

[illegible]

Print at 100% / Actual size — do NOT hit to page. Two-sided, flip on the SHOK! edge. The numeral on both faces is your check: cut ONL card first. This bar must measure exactly 80 mm.